Name _____ Period _____ Date _____

5-6 Lesson Quiz: Inverse Relations and Functions

short form

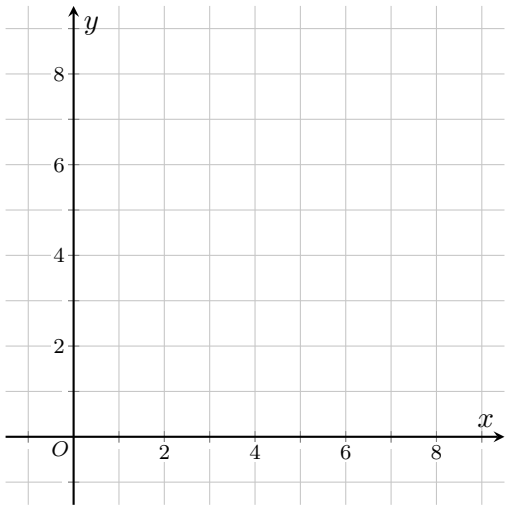
Question numbers match the Savvas Lesson Quiz (questions 1 and 4 are not on this form).

2. Sketch the graph of f and the inverse of f, if $f(x) = \frac{1}{2}x + 3$.

Show your work: switch x and y and solve for y (or swap the coordinates of two points of f).

$y = \frac{1}{2}x + 3 \longrightarrow x =$ _____

$f^{-1}(x) =$ _____



3. Let $f(x) = x^2 - 2x + 1$. Find the inverse function of f by identifying an appropriate restriction of its domain.

Ⓐ The inverse is $f^{-1}(x) = \sqrt{x+1}$ if the domain of f is restricted to $x \geq -1$.

Ⓑ The inverse is $f^{-1}(x) = \sqrt{x} + 1$ if the domain of f is restricted to $x \geq 1$.

Ⓒ The inverse is $f^{-1}(x) = \sqrt{x} - 1$ if the domain of f is restricted to $x \geq 0$.

Ⓓ The function does not have an inverse.

Show your work. Write f as a square, name the vertex, then switch x and y:

5. Let $f(x) = 2x + 4$ and $g(x) = \frac{1}{2}x - 2$. Complete the sentence. Check **one** box.

The function $g(x) = \frac{1}{2}x - 2$ _____ $f(x) = 2x + 4$.

☐ is the inverse of ☐ is equivalent to ☐ is the opposite of ☐ has the same intercepts as

Scoring: Question 2 and Question 3 are 10 points each. Question 5 is a bonus of 5 points. Total out of 20.

5-6 Lesson Quiz, short form: answer key

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2. Graph of f and f^{-1}. f is the line through $(0, 3)$ with slope $\frac{1}{2}$ (also $(2, 4)$, $(6, 6)$). Switch and solve: $x = \frac{1}{2}y + 3$, $2(x - 3) = y$, so $f^{-1}(x) = 2x - 6$: the line through $(3, 0)$ with slope 2 (also $(4, 2)$, $(6, 6)$). The two lines are reflections across $y = x$ (for the teacher: they cross at $(6, 6)$ on $y = x$; students are not asked for it). TE key: graph shown (two lines, shallow f with y-intercept 3, steep inverse with x-intercept 3, crossing at $(6, 6)$). Savvas quiz item 2, DOK 2, skill F65. **10 points:** 4 for the graph of f, 4 for the graph of the inverse, 2 for the work shown (the equation $f^{-1}(x) = 2x - 6$, or two points of f with their coordinates swapped).

3. B. $f(x) = x^2 - 2x + 1 = (x - 1)^2$, vertex $(1, 0)$. Restrict to $x \geq 1$; then $x = (y - 1)^2$, $\sqrt{x} = y - 1$, $f^{-1}(x) = \sqrt{x} + 1$, $x \geq 0$. Distractor A is the inverse of $x^2 - 1$ and C of $(x + 1)^2$: each starts from the wrong vertex. Savvas quiz item 3, DOK 2, skill F65. **10 points:** 6 for B, 4 for the work (the square $(x - 1)^2$ or the vertex, and the switch).

5. is the inverse of. $x = 2y + 4$ gives $y = \frac{1}{2}x - 2 = g(x)$. Check: $f(g(x)) = 2(\frac{1}{2}x - 2) + 4 = x$. Savvas quiz item 5, DOK 2, skill F65. **Bonus 5:** the inverse box and no other.

Items left off this form (available for a retake or Form B): Savvas item 1, missing values in the table of the inverse (answer $a = -1$, $b = 1$, $c = 3$, $d = -2$; DOK 1, F65; the packet's "switch the columns" table rehearses it); Savvas item 4, inverse of $f(x) = \frac{1}{2}\sqrt[3]{x - 3} + 5$ in the form $f^{-1}(x) = __ + __(x - __)^3$ (answer 3; 8; 5, that is $f^{-1}(x) = 3 + 8(x - 5)^3$; DOK 2, F65; taught by Example 4, cut from the fast packet).

IXL replacement: SmartScore 70 or above on ZRQ, K2W and 9KT replaces this quiz with 20/20.